

StoryCube: Tangible Prompt Engineering for Pre-Literate Children

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Figure 1: StoryCube enables children to prompt generative AI to create short stories using tangible tokens that magnetically attach to the cube. (Left) A preschooler places tokens on StoryCube during use. (Center) The StoryCube device with attached tokens. (Right) The collection of tangible story tokens used to compose prompts.

Abstract

Preschool children are largely excluded from interacting with LLMs because current prompt interfaces rely on fluent speech, writing, or reading. As generative AI increasingly pervades children’s everyday environments, they need age-appropriate, accessible ways to express intent and gain agency in AI-mediated experiences. We introduce StoryCube, a tangible prompt engineering interface that lets children compose structured prompts using physical story tokens to generate audio stories. StoryCube is novel in enabling structured prompting through tangible interaction, supporting pre-literacy access and collaborative, hands-on prompt composition rather than text- or voice-based prompting. In a preliminary study with six children (ages 5–8) and two educators, children used StoryCube intuitively with minimal guidance, explored diverse token combinations, and reported high enjoyment, while also noting mixed story quality. Findings suggest that tangible prompt engineering is a promising concept for child-centered AI interaction, motivating further research on parametrization, token grammar, safety, and educational integration.

CCS Concepts

• **Human-centered computing** → *Empirical studies in interaction design*; **Interaction paradigms**;

Keywords

tangible user interfaces, prompt engineering, large language models, children, storytelling, audio interfaces

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1 Introduction and Related Works

Large Language Models (LLMs) are now widely available, but most interaction paradigms assume users can read, type, and navigate screen-based interfaces. This creates a fundamental barrier for pre-literate children. As a result, children’s LLM use often becomes mediated: adults help reformulate questions and the children’s intent into prompts and interpret model outputs [4]. While the latest generative artificial intelligence (AI) chatbots support voice-based conversation modes, children in early childhood often struggle to interact with these interfaces, due to the quality of speech-to-text transcription and interaction timings [12].

Prior work suggests that tangible user interfaces (TUIs) can lower entry barriers by leveraging familiar physical interaction, supporting playful exploration, and fostering collaboration in learning contexts. Storytelling is a particularly strong application domain for TUIs in early literacy education [10]. Commercial products such as the Toniebox¹ already demonstrate the appeal of tangible, audio-first story consumption, but offer only fixed content without children-initiated variation. Recent research has also started exploring how generative AI can support interactive story experiences



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¹<https://us.tonies.com/>

for children and families, for example, through LLM-based story reading support [2] and parent-oriented human-AI collaborative storytelling chatbots [15].

However, these approaches remain largely screen- or voice-based and do not enable children to shape the model’s behavior through hands-on prompt construction. In contrast, research prototypes like StoryMat [9] and tangible storytelling systems using cubes and tabletop interaction [1, 6] show how children can shape narratives through embodied interaction, yet they do not address the emerging opportunity of enabling children to steer generative AI behavior through prompt construction.

In LLM research, prompt engineering typically refers to the deliberate design of structured textual instructions to steer model behavior without modifying model weights [5]. This practice has become a key mechanism for shaping LLM behavior and is usually performed by developers or advanced users and often involves iterative refinement of constraints and instructions [7, 13]. Research has begun exploring interaction paradigms beyond current chat applications that translate user actions into engineered prompts, including direct manipulation and graphical interactions [3, 8]. The potential of tangible prompt engineering, especially for children, remains underexplored despite its promise for accessibility and engagement. At the same time, children’s interaction with digital technology is shaped by long-standing societal and parental concerns, which extend to generative AI around age-appropriateness, hallucinated misinformation, and responsible introduction with guidance [11, 14].

To address this gap, we introduce StoryCube, a tangible prompt engineering interface that lets pre-literate children compose structured LLM prompts using physical story tokens. We reinterpret prompt engineering as a design space for tangible, parameterized interaction with generative models, where prompt structure is pre-engineered, and children manipulate high-level semantic components. Instead of typing or speaking a prompt, children select and attach tangible story elements (e.g., characters, locations, plots) to a cube, which then constructs a structured prompt and generates a short audio story reflecting their choices. StoryCube builds on the familiarity of tangible, audio-first storytelling while extending it toward generative variability and child-driven *prompting* as a hands-on, collaborative, and playful activity.

This work contributes (1) a fully working prototype for tangible prompt engineering with pre-literate children, and (2) a preliminary mixed-methods evaluation with six children (aged 5–8) in a daycare setting, providing initial insights into usability, engagement, and emergent collaboration patterns. Our findings suggest that tangible prompt engineering is a promising direction for making LLM interaction more accessible to pre-literate children, yet requires an open discussion within children’s play and early childhood settings regarding appropriate AI guardrails and safety mechanisms, bias and representation in generated narratives, children’s creativity, and potential long-term effects on play practices and dependency.

2 StoryCube

StoryCube was designed with four core objectives in mind, providing children with direct control over the prompt-generation process. Our goal was to support pre-literacy usability by enabling

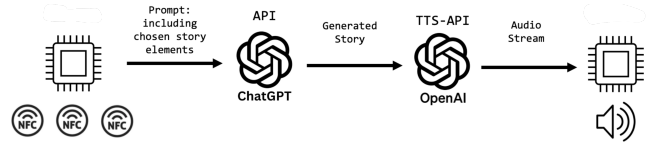


Figure 2: StoryCube end-to-end generation pipeline. Tangible story tokens are read via NFC, assembled in a structured prompt, and sent to the ChatGPT API to generate a short story. The generated text is then converted to speech via the OpenAI TTS API and streamed back to the device for audio playback.

prompt construction without typing skills. We minimize interaction complexity by constraining input to tangible object placement and a single action-button paired with salient auditory feedback. Collaborative use is facilitated by enabling concurrent engagement by multiple children without introducing a bottleneck in the interaction flow. Finally, child-appropriate experiences are ensured by producing narratives in age-appropriate, simple language and design decisions oriented toward inclusivity and the mitigation of LLM biases.

Interaction Concept. Children can interact with StoryCube by selecting up to four tangible tokens and placing them on dedicated reading areas. Each placement triggers an audio cue that announces the selected element (e.g., tiger), ensuring that children can confirm their selection. When the button is pressed, the system generates an audio story based on the selection. During playback, pressing the button pauses and resumes. If tokens are changed, pressing the button again triggers a new story generation based on the updated token set.

Token Language and Prompt Structure. Currently, our StoryCube catalog provides 19 different tangibles categorized into four color-coded groups (cf. Figure 1 right): subjects (e.g., astronaut, dinosaur, tiger), plots (e.g., adventure, dream, mystery), locations (e.g., forest, mountain), and adjectives (e.g., fearless, smart). These tokens map into a structured prompt that integrates the selected words into a story request. The story generation also relies on a fixed system prompt that instructs child-friendly language, clear narrative structure, and balanced gender representation among characters.

3 Implementation

StoryCube is implemented as a standalone, USB-powered 3D printed cube centered around a Raspberry Pi 5. The computing unit is connected to four RFID modules for reading NFC tags, a speaker for audio output, and a large single generate/play/pause button. An analog volume dial enables volume control. Each tangible consists of an NFC tag embedded in a durable housing with a high-contrast symbol. Symbols are slightly raised to improve haptic recognition. The device is designed to be robust for children’s use. Story tokens lock into one of four slots and are held in place using neodymium magnets.

A Python script orchestrates the full interaction loop: it monitors NFC reads, tracks selected tangibles, constructs the structured

prompt, and calls the OpenAI API to generate the story. The generated text is converted into speech via the OpenAI text-to-speech API, and audio is streamed in real time to reduce latency between pressing the button and hearing the story.

Although local deployment would be desirable from a cost and infrastructure perspective, we selected a commercial API after evaluating open-source alternatives, including smaller models that could run directly on the Raspberry Pi. In internal tests, these options did not provide sufficient narrative coherence, linguistic quality, and naturalness of voice. For our study, we therefore prioritized the quality of LLM output to provide a compelling experience.

To ensure stories remain appropriate and coherent for young audiences, StoryCube uses a predefined system prompt that instructs the model to produce child-friendly stories in simple language. The prompt specifies length, story structure, and output formatting. It also includes an instruction to balance and diversify the gender distribution among characters. In the current prototype, the generated story is not post-processed before text-to-speech conversion.

4 Preliminary Study

We conducted a preliminary study to assess whether pre-literate children can use tangible prompt engineering intuitively and how the system supports engaged, collaborative exploration. The deployment took place in a daycare setting with educator supervision. The study was reviewed and approved by our university’s ethics board. Written permission was also obtained from the daycare leadership. Parents or guardians provided written consent for children’s participation, and an educator was present throughout all sessions.

4.1 Method

We conducted a preliminary mixed-methods study across three consecutive days in an elementary school classroom during an after-school care program. Each session lasted approximately 30 minutes of free exploration with StoryCube in a familiar environment, followed by group interviews after a short break.

We collected (1) semi-structured group interviews with children and educators (audio-recorded, transcribed, and manually corrected), (2) in-situ observations during free exploration, and (3) system logs capturing interaction events. Interview transcripts were analyzed via open coding in MAXQDA.

4.2 Participants and Procedure

Six children (ages 5–8, median 6.5; 2 male, 4 female) participated. All were native German speakers, and none had prior experience with the tangible storytelling system. Children participated in different group constellations throughout the days (day 1: C1–C3, day 2: C1, C2, C4–C6, day 3: C1, C6). Two educators were interviewed: one leisure educator and one German teacher focused on language support.

Each day, children received a short introduction, then used StoryCube in a 30-minute free exploration session on the classroom floor while we collected observations and system logs (cf. Figure 3). One researcher remained present during use and intervened only when necessary. After a short break, we conducted semi-structured group interviews with children and educators separately.



Figure 3: Preliminary daycare session with StoryCube. Children gathered around the tangible interface to collaboratively select story tokens and listen to the generated story.

5 Results

Our qualitative analysis of interview transcripts and in-situ observations used open coding to capture children’s and educators’ experiences with StoryCube. Besides quantitative results and educators’ perspectives, we identified four themes in children’s feedback and behavior: (1) engagement and enjoyment, (2) usability and independent use, (3) perceived story quality, and (4) collaboration and creative exploration:

Engagement and Enjoyment. Across sessions, children described StoryCube as fun and expressed a strong desire to use it again. One child (C5) said they would use it as a bedtime story player. Observations backed this enthusiasm: children gathered around the cube, were excited by the large red start button, and often wanted to press it together. Some children who participated in multiple sessions asked immediately whether StoryCube would be available again, suggesting sustained interest beyond a single exposure.

Usability and Independent Use. Children successfully operated StoryCube on the first try. No usability challenges were observed during initial interaction. Children were able to attach tokens, trigger story generation, and iterate on token combinations without requiring step-by-step guidance. One child summarized the interaction as “just token in and go,” (C3) and another said: “I just put the token in, and it works.” (C2) No major usability breakdowns were observed, and children did not require adult assistance during use.

Story Quality. Children appreciated that stories were different each time and that generated content reflected their choices (e.g., referencing a “clever dinosaur” corresponding to selected tokens). At the same time, some children wanted more dramatic plot events or more varied narration, and they noticed repeated character names across different stories.

Collaboration and Creative Exploration. In group sessions, children naturally collaborated by negotiating token choices, adopting turn-taking strategies (each child selects one token), and interacting in parallel. While one child attached tokens, others explored options. Children also engaged in playful exploration of the color-coded categories, including experimenting with single-color (category) combinations to see what stories would result.

Quantitative Results. Across three sessions (30 minutes each), the system recorded 30 initiated story-generation processes. Of these, 22 reached the stage of audio playback initiation; 10 were completed to the end, and 16 were explicitly cancelled. All 19 tangibles were used at least once. The most frequently selected tokens were *magical*, *adventure*, and *tiger*. Across the 30 initiations, children produced 23 unique token combinations. A separate log analysis of generated short stories revealed repetition in the model's naming choices: "Mia" appeared in 13 stories, aligning with children's qualitative feedback on repeated names.

Educators' Perspectives. Both educators expressed cautious optimism about AI in entertainment and education, especially for language learning, while emphasizing a responsible, age-appropriate introduction. One educator highlighted that a future StoryCube being capable of adapting language level and complexity could support children with an immigrant background and be *truly helpful* in resource-constrained daycare contexts. Another warned against overreliance, stressing the need for guidance and framing AI as a tool that can be powerful but also *risky* without supervision.

6 Discussion

StoryCube reframes prompt engineering as a tangible composition task. Rather than relying on typed language, children communicate intent through physical tokens and receive immediate, comprehensible feedback via speech. This creates a low-barrier pathway to engage with generative AI as an entertaining, interactive medium and not merely a static content source.

A notable outcome was how easily StoryCube supported collaborative play. The shared physical interaction space enabled parallel actions (choosing, attaching, planning) that would be difficult to replicate with a single touchscreen or text input.

Children valued that stories were always different and did not become repetitive like fixed-content storytelling devices. However, mixed feedback on excitement, voices, and repeated names highlights a key design tension: enabling children's interactional and expressive agency through open-ended generation while maintaining consistently engaging and child-appropriate story quality. This tension motivates future design questions: *How should token constrain (or expand) the space of possible outputs? How can systems maintain predictability and safety while preserving the delight of surprise?*

One educator identified great language-learning potential, especially if story complexity could be aligned to children's language level. This might be of great interest to children learning a second language. This highlights a direction where tangible prompt engineering becomes more than entertainment: it can promote comprehension and vocabulary exposure. At the same time, educators' caution underscores the need for carefully designed safeguards,

transparency about generative limitations, and mediated introduction strategies.

6.1 Limitations and Future Work

This work is preliminary. The study involved a small sample and short sessions in a children's daycare context. Effects may not generalize to home use or longer-term engagement. The novelty effect may have influenced enthusiasm, and the presence of a researcher and educator may have shaped interaction behavior. Long-term home deployments may yield different usage patterns, collaboration dynamics, and expectations toward story quality and reliability. Nevertheless, these early findings provide a promising foundation for designing tangible, child-centered prompt interfaces and motivate broader, longer-term evaluations in everyday settings.

System-wise, the current token set constrains expressiveness, and story generation quality is not yet consistently aligned with children's expectations. Future work will therefore focus on personalization, language proficiency adaptation, and dynamic storytelling where token changes during playback steer the narrative in real time and longitudinal home deployments to capture sustained engagement and appropriation over time.

7 Conclusion

We presented StoryCube, a tangible user interface that makes LLM interaction accessible to pre-literate children by transforming prompt engineering into a physical, collaborative process of story composition. A preliminary study indicates that children can use the system, find it engaging, and naturally collaborate around it. Our findings motivate tangible prompt engineering as a promising design space for child-centered AI interaction, while opening a broad discussion around agency, safety, quality control, and educational integration for generative AI in children's everyday environments.

AI Disclosure

OpenAI ChatGPT 5.2 was used during manuscript preparation for language polishing and structural suggestions. The intellectual contribution remains with the authors. OpenAI ChatGPT-4o was used as part of the StoryCube system to generate and narrate stories.

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